

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix.

OUR WORLD IN DATA

75% of Global GHG Emissions Still Come from Fossil Fuels

Since the Industrial Revolution, energy systems have been dominated by fossil fuels, creating a dual crisis of climate change and air pollution.

The Status Quo

75%

of global greenhouse gas emissions are driven by fossil fuel combustion.

The Health Impact

5M+

premature deaths annually attributed to local air pollution from fossil fuels.

The Mandate

Shift

The world must rapidly transition toward low-carbon energy technologies.

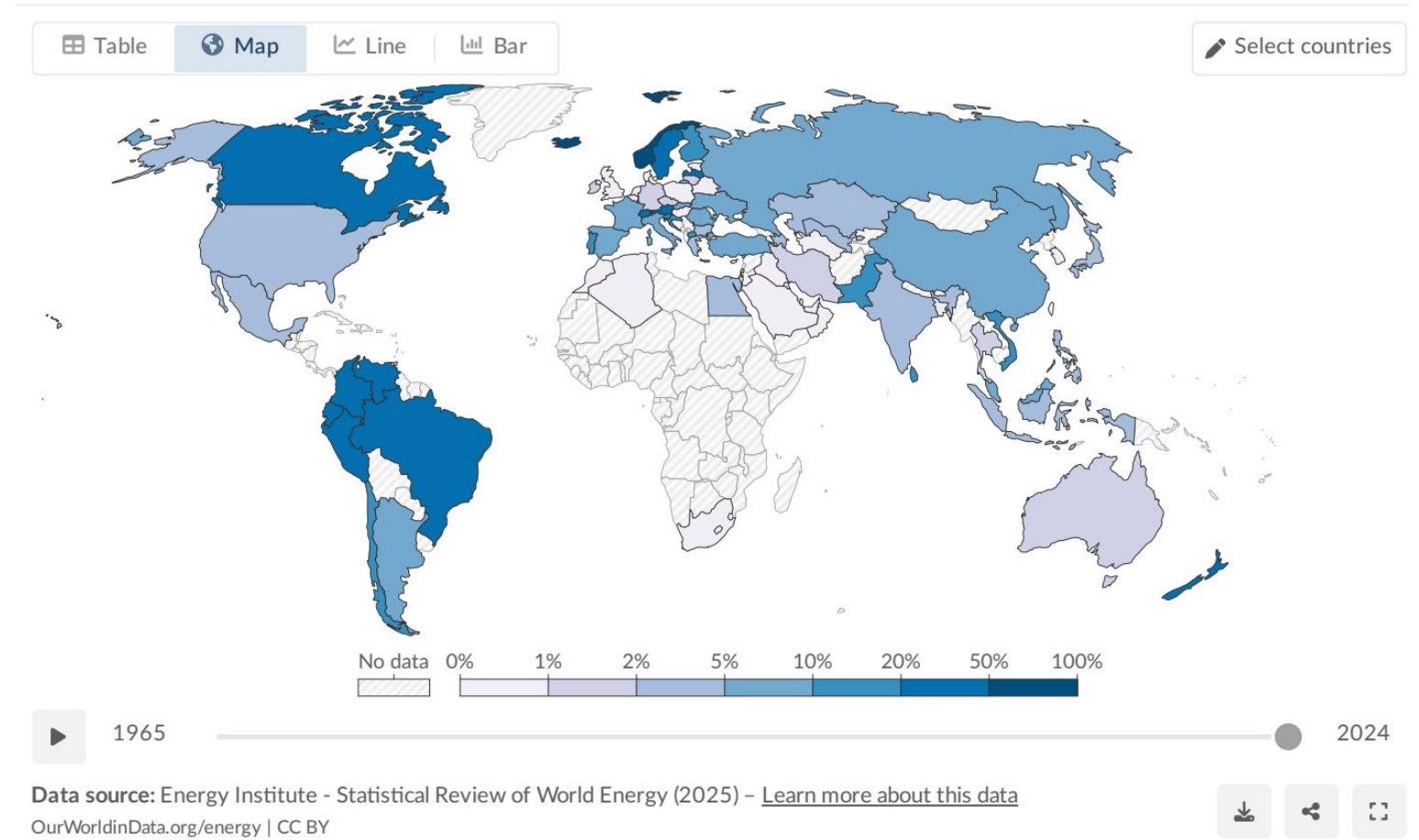
Renewables Now Supply ~14% of Global Primary Energy

The Primary Energy Mix

Approximately one-seventh (~14%) of the world's primary energy now comes from renewable technologies (2024 data, substitution method).

Why is this lower than electricity?

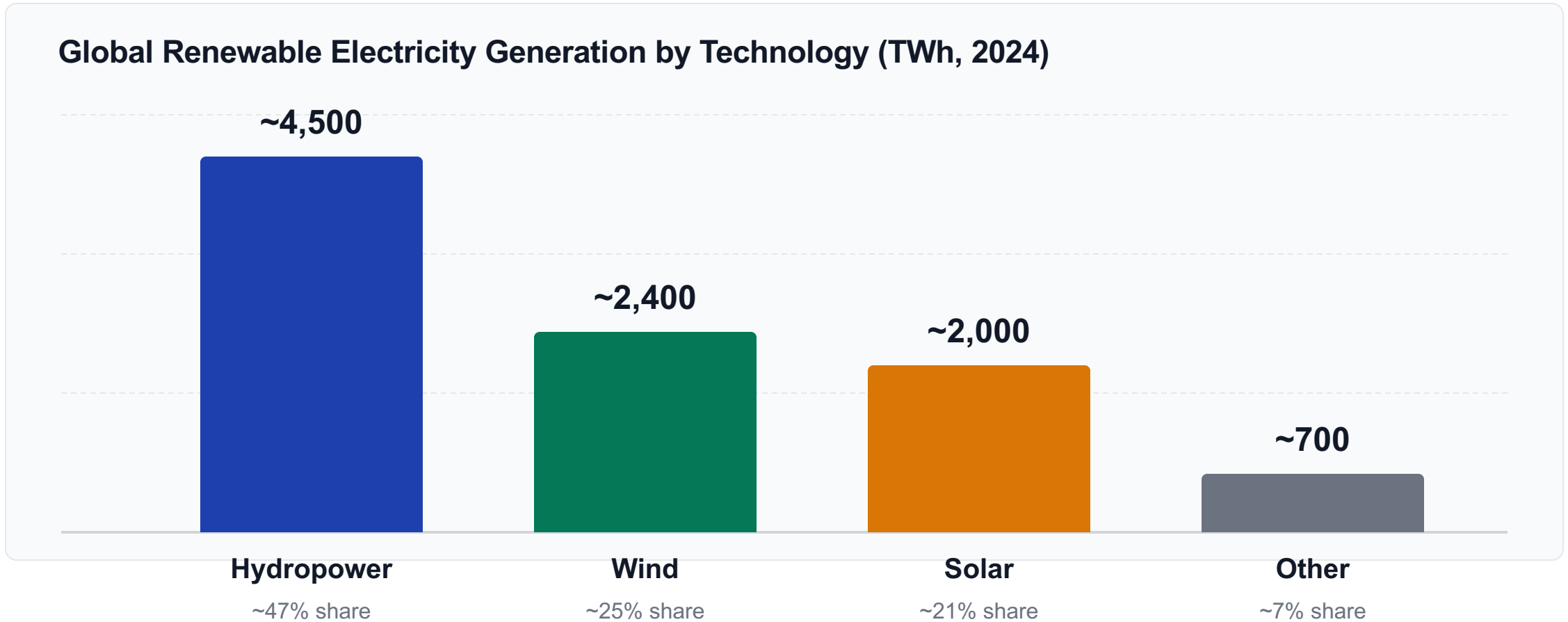
Primary energy encompasses electricity, transport, and heating combined. Transport and heating are harder to decarbonize and remain highly reliant on oil and gas.



Share of primary energy from renewables by country, 2024

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

While hydropower still leads, wind and solar combined now nearly match hydro's output and are growing much faster.



Almost One-Third of Global Electricity Now Comes from Renewables

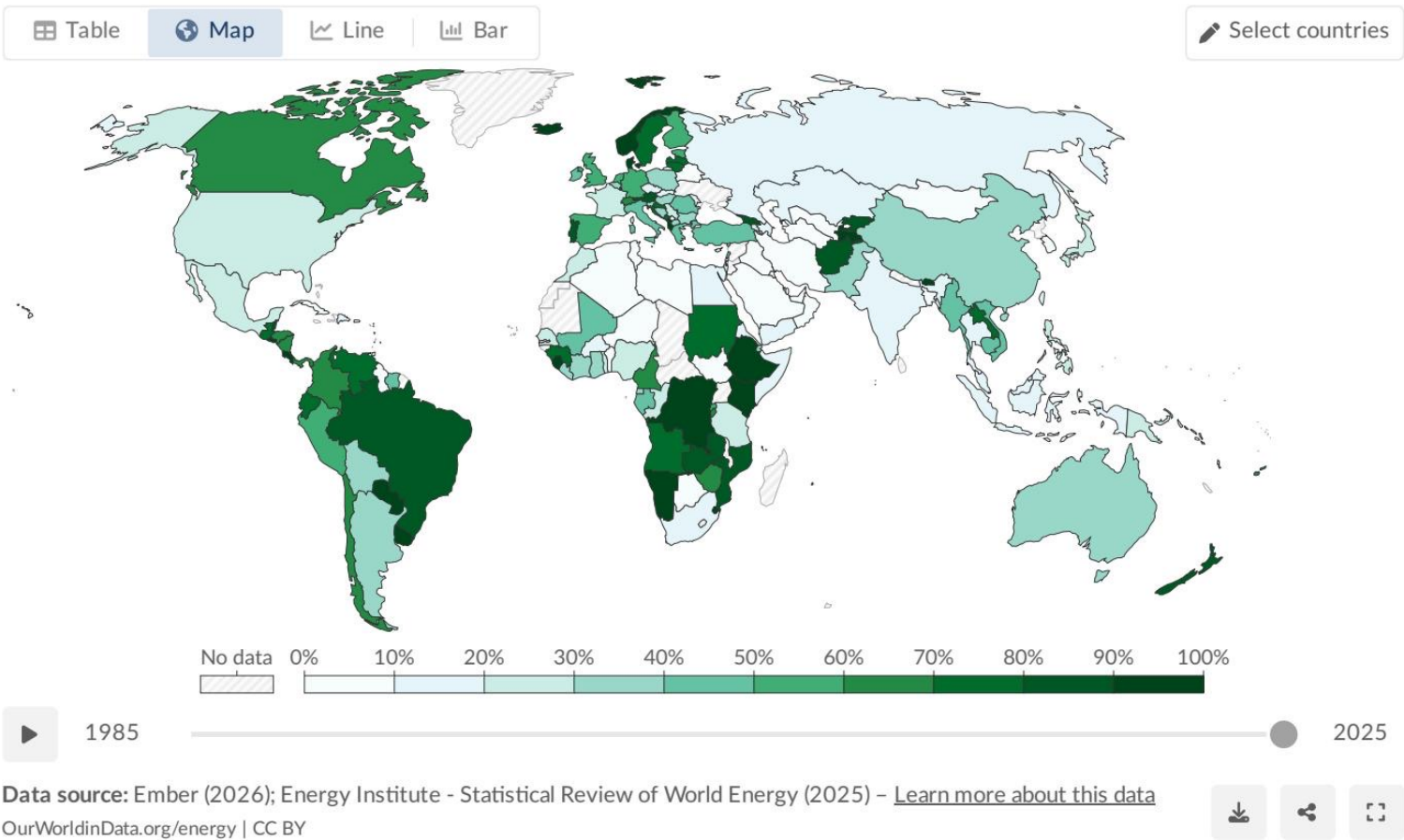
Regional Leaders vs Laggards

Globally, almost one-third (~30%) of electricity now comes from renewable sources (2025). This is substantially higher than the primary energy share.

Selected Country	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.



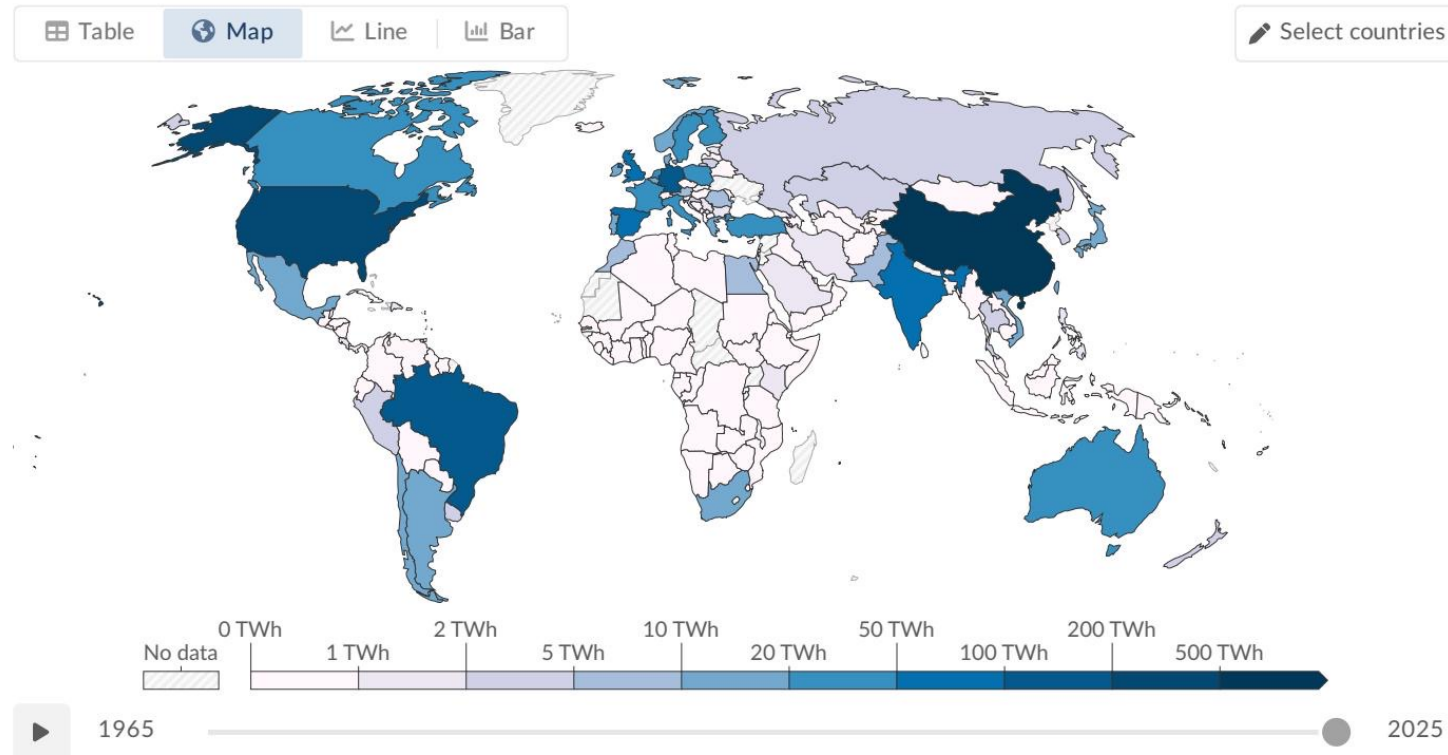
Share of electricity from renewables by country, 2025

Hydropower Remains the Largest Renewable Source at ~50% of Generation

The Foundation of Renewable Electricity

Hydropower remains the single largest renewable electricity source, accounting for roughly half of all renewable generation. It has been generating electricity at scale for over a century.

Top Producers
China, Brazil, Canada



Hydropower generation by country, 2025

Wind and Solar Are the Fastest-Growing Energy Technologies in History

The Growth Engine

Solar energy has seen dramatic growth since 2010, driven by falling costs and policy support, making it the fastest-growing source globally.

Wind Generation

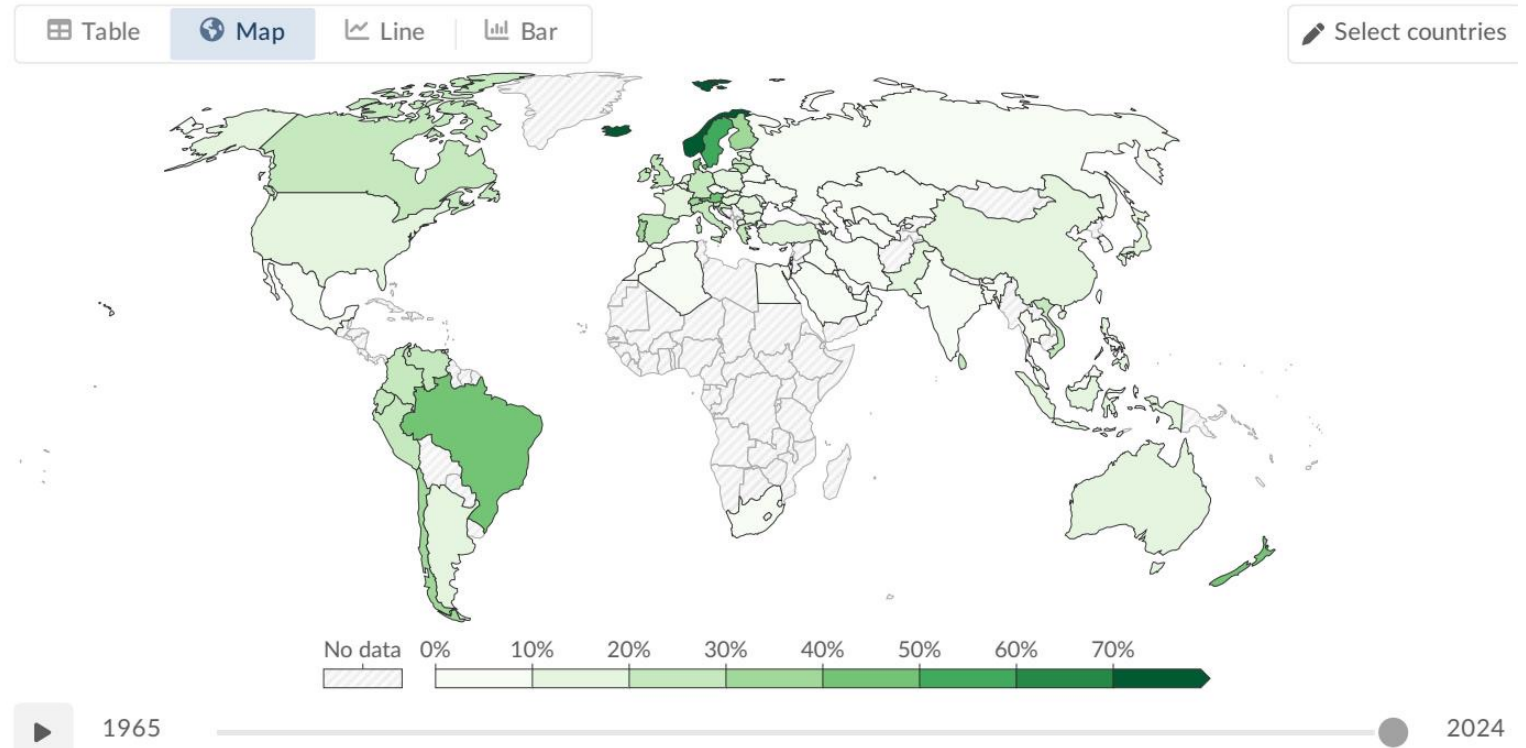
~2,400TWh

~25%
of renewable

Solar Generation

~2,000TWh

~21%
of renewable



Wind power generation by country, 2025

Wind and solar combined now nearly match hydropower output.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lea

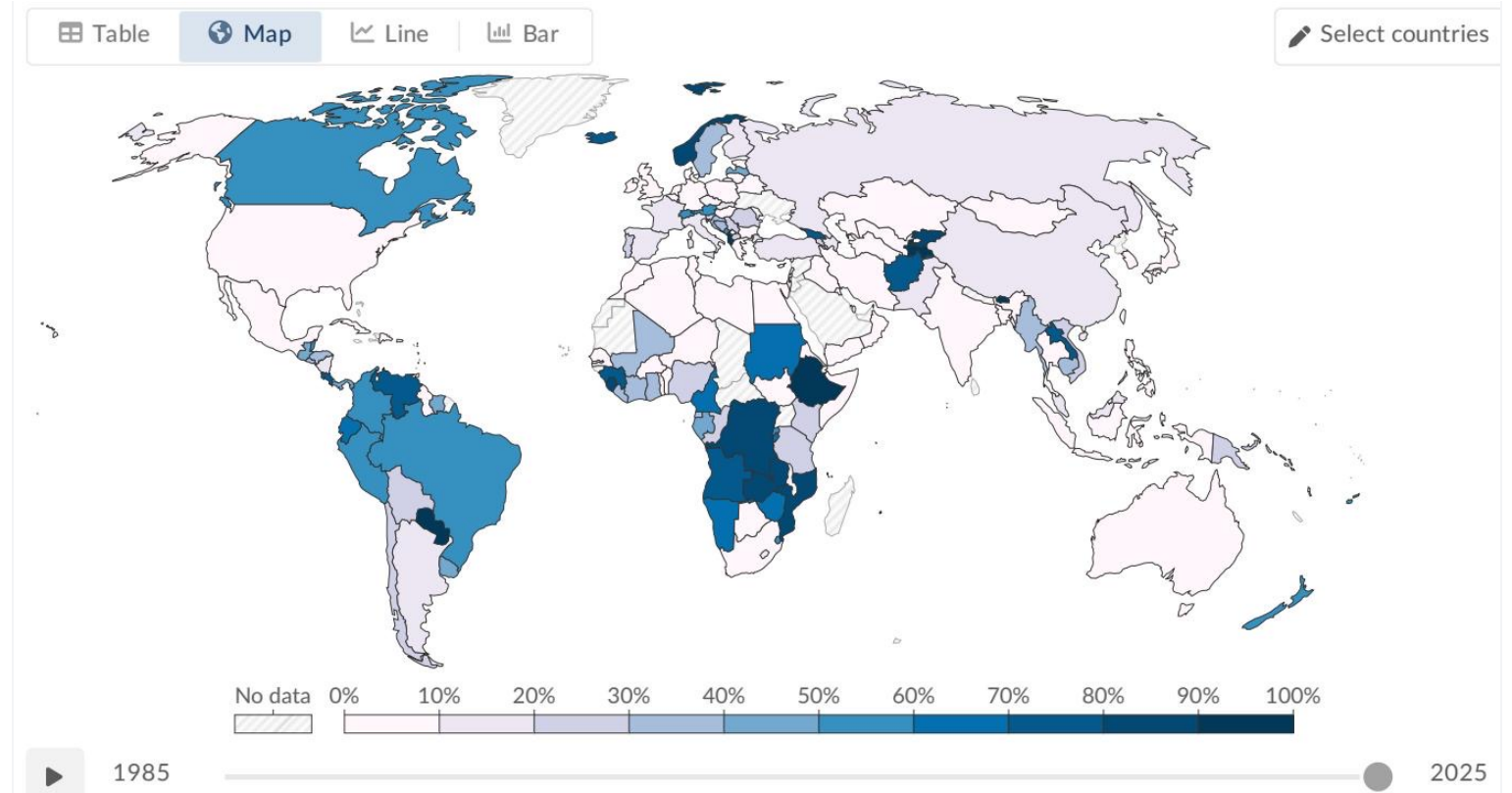
An Uneven Transition

Leaders (65% - 98% Renewable Share)

Scandinavian countries, Brazil, and Canada benefit from abundant hydropower resources. Denmark and Germany have aggressively invested in scaling wind and solar generation.

Laggards (<10% Renewable Share)

Much of Sub-Saharan Africa currently lacks minimal renewable electricity infrastructure. Many Middle Eastern nations remain almost entirely dependent on fossil fuels.



Share of electricity from hydropower by country, 2025

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

- **Current Milestone: ~14% of global primary energy and ~30% of global electricity**
now come from renewable sources.
- **Hydropower remains dominant:**
Generating roughly half of all renewable electricity, though growth potential is limited compared to wind and solar.
- **Wind and solar are the fastest-growing energy technologies in history:**
Combined, they now produce nearly as much electricity as hydropower.
- **Stark regional disparities persist:**
Scandinavia and Latin America lead, while much of the Middle East and Sub-Saharan Africa remain below 10%.

The Path Forward: To meet climate targets, the transition pace across heating and transport must accelerate.